

BABUSHKIN, Russian Federation

WMO: 308220

Lat: 51.708N

Lon: 105.852E

Elev: 468

StdP: 95.83

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-26.2	-23.4	-29.1	0.3	-25.7	-26.3	0.4	-22.7	10.5	-15.6	9.9	-15.1	1.7	240	0.694

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.1	22.7	16.7	21.1	16.2	19.8	15.7	18.0	21.6	17.1	20.0	16.3	19.0	2.5	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.7	12.6	19.8	15.9	11.9	18.9	15.0	11.3	18.1	53.0	21.7	50.2	20.1	47.6	19.1	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 10.4	9.5	8.4	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	-27.7	19.9	2.8	2.3	-29.8	21.5	-31.4	22.9	-33.0	24.2	-35.1	25.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.3	-13.2	-14.5	-7.3	1.1	6.6	11.9	16.0	15.3	9.6	2.7	-4.5	-8.6
	DBStd	11.07	5.11	5.38	5.65	3.39	3.20	2.76	2.54	2.45	3.07	3.70	4.59	4.23
	HDD10.0	3618	719	686	537	268	114	12	0	1	43	227	434	577
	HDD18.3	6213	978	919	796	518	365	194	81	97	261	484	684	835
	CDD10.0	458	0	0	0	0	7	68	185	165	32	1	0	0
	CDD18.3	11	0	0	0	0	0	0	7	3	0	0	0	0
	CDH23.3	34	0	0	0	0	1	1	22	9	1	0	0	0
	CDH26.7	4	0	0	0	0	0	0	3	1	0	0	0	0
(14) Wind	WSAvg	3.3	3.5	2.7	2.8	3.2	3.5	3.0	2.6	2.8	3.4	3.5	4.2	4.5
(15) Precipitation	PrecAvg	522	17	8	11	25	33	69	112	90	72	29	25	27
	PrecMax	720	40	22	32	58	66	205	304	216	188	83	54	64
	PrecMin	375	2	0	2	1	2	6	16	18	17	5	5	5
	PrecStd	91	9	5	8	15	16	45	65	42	39	19	11	15
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.1	-0.1	8.0	14.8	20.9	22.5	25.7	24.3	20.6	14.6	7.0	2.3
		MCWB	-1.5	-3.0	2.4	6.4	10.9	14.9	18.5	17.3	14.1	8.5	3.5	-0.7
	2%	DB	-2.3	-2.5	4.6	10.4	16.6	20.0	23.2	22.2	17.6	11.2	5.2	0.3
		MCWB	-3.7	-4.4	1.0	4.7	8.8	14.3	17.7	16.3	12.4	7.1	2.1	-1.8
	5%	DB	-4.1	-4.6	2.4	8.0	14.1	18.2	21.5	20.6	15.8	9.4	3.2	-1.0
		MCWB	-5.5	-6.2	-0.3	3.4	8.1	13.3	17.1	16.2	11.8	6.0	0.7	-2.8
	10%	DB	-6.3	-6.7	0.8	6.2	12.2	16.7	20.2	19.3	14.3	8.0	1.8	-2.6
		MCWB	-7.5	-7.9	-1.5	2.5	7.0	12.6	16.4	15.7	11.1	5.0	-0.6	-4.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.5	-2.5	3.1	7.3	12.1	16.8	19.5	18.7	15.7	9.0	4.1	-0.3
		MCDB	0.0	-0.6	6.2	13.0	18.7	19.9	22.9	21.9	18.7	13.4	6.8	1.6
	2%	WB	-3.5	-4.3	1.4	5.3	10.0	14.9	18.5	17.5	13.7	7.7	2.4	-1.5
		MCDB	-2.2	-2.6	4.4	9.8	14.5	18.5	21.9	20.6	16.5	10.9	4.7	0.2
	5%	WB	-5.5	-6.2	0.0	3.8	8.7	13.9	17.7	16.8	12.4	6.4	0.9	-2.8
		MCDB	-4.2	-4.7	2.1	7.5	13.3	17.2	21.1	19.5	14.8	9.0	2.9	-1.1
	10%	WB	-7.4	-7.9	-1.4	2.7	7.5	13.0	16.8	16.1	11.4	5.2	-0.4	-4.2
		MCDB	-6.3	-6.7	0.7	5.9	11.9	16.1	19.6	18.8	13.9	7.7	1.6	-2.6
(35) Mean Daily Temperature Range	5% DB	MDBR	7.0	8.3	8.8	7.4	7.8	7.3	6.1	5.9	6.5	6.5	6.1	5.9
		MCDBR	7.8	10.0	10.6	11.1	12.1	10.0	8.7	8.4	9.1	8.6	7.0	6.8
		MCWBR	6.7	8.5	7.5	6.3	6.3	5.6	4.6	4.5	5.4	6.1	5.4	5.8
	5% WB	MCDBR	7.5	9.9	9.3	10.4	10.6	8.5	7.4	7.1	8.0	8.1	6.7	6.6
		MCWBR	6.6	8.5	6.7	6.3	6.1	5.4	4.4	4.3	5.3	6.0	5.5	5.8
(40) Clear-Sky Solar Irradiance	taub		0.269	0.295	0.331	0.378	0.442	0.445	0.434	0.394	0.340	0.316	0.288	0.270
	taud		2.120	2.104	2.146	2.236	2.089	2.199	2.278	2.375	2.473	2.432	2.207	2.090
	Ebn at Noon		721	810	853	856	812	814	817	834	848	797	694	640
	Edh at Noon		74	101	120	124	152	137	125	108	87	74	68	65
(44) All-Sky Solar Radiation	RadAvg		1.28	2.34	3.70	4.75	5.68	5.92	5.48	4.62	3.58	2.24	1.27	0.90
	RadStd		0.06	0.12	0.18	0.30	0.31	0.34	0.36	0.26	0.34	0.16	0.09	0.07

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page